

MHS Senior 5 Pure Maths Revision Questions 2020

1. Simplify: $\frac{3(2^{n+1}) - 4(2^{n-1})}{2^{n+1} - 2^n}$.
2. Solve the indicial equation: $9^x - 10(3^{x-1}) + 1 = 0$.
3. Express $\frac{1 + \tan 30^\circ}{1 - \tan 30^\circ}$ in the form $a + b\sqrt{c}$.
4. Solve and verify the answer: $\sqrt{2x+3} - \sqrt{x-2} = \sqrt{x+1}$.
5. If $p = \log_2 x$ and $q = \log_{12} x$, show that $\log_6 2 = \frac{q}{p-q}$.
6. If $a = \log_b c$, $b = \log_c a$ and $c = \log_a b$, prove that $abc = 1$.
7. If $a^2 + b^2 = 23ab$, show that $\log a + \log b = 2 \log(\frac{a+b}{5})$.
8. If $x = \log_a bc$, $y = \log_b ac$ and $z = \log_c ab$, show that $x + y + z = xyz - 2$.
9. Solve the equation: $\log_3 x = \log_6 4x$
10. Solve the logarithmic equation: $\log_n 5 + 4 \log_5 n = 4$.
11. Solve the equation $\log_3 x + \log_9 x + \log_{81} x = \frac{21}{4}$.
12. Solve the equation $\log_3 x = \log_9(1 - 3x^2)$.
13. Express $p(x) = 3x^2 - 6x + 11$ in the form $a(x+b)^2 + c$. Hence deduce the least value of $p(x)$.
14. One root of $x^2 + 3x + p = 0$ is q times the other. Prove that $p(1+q)^2 = 9q$.
15. If $x^2 + bx + c = 0$ and $x^2 - px + q = 0$ have a common root, show that $(c-q)^2 = (b-p)(cp-bq)$.
16. $x^2 - 13x + 36 = 0$ has roots α and β . Find the quadratic equation with roots $\sqrt{\alpha}$ and $\sqrt{\beta}$.
17. Find the range of values of k for which $4x^2 - 5x - k = 0$ has real roots.
18. If $y = mx + c$ is a tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, prove that $c^2 = a^2m^2 + b^2$.
19. Find the intersection of the curve $xy = 9$ and the line $x + y = 6$.
20. Prove that $\sin^2 \theta (1 + \sec^2 \theta) = \sec^2 \theta - \cos^2 \theta$.
21. Prove that $\sec \theta \operatorname{cosec} \theta = \cot \theta \sec^2 \theta$.
22. Prove that $\sec^4 \theta - \tan^4 \theta = 1 + 2 \tan^2 \theta$.
23. If $p = \sin \theta$ and $q = 4 \cot \theta$, show that $p^2 q^2 = 16(1 - p^2)$.
24. Solve $\operatorname{cosec} \left(\frac{\theta}{3}\right) = \frac{2}{\sqrt{3}}$ for $0 \leq \theta \leq 2\pi$.
25. Solve $\sqrt{3} \sec x + 2 = 0$ for $0^\circ \leq x \leq 360^\circ$.
26. Solve: $2 \sin^2 \theta = 3(1 - \cos \theta)$ for $0 \leq \theta \leq 2\pi$.
27. Solve: $8 \sin^4 \theta - 10 \sin^2 \theta + 3 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.